

3 Geometric and negative binomial distributions	3.1	Geometric and negative binomial distributions.	Models leading to the distributions $p(x) = p(1 - p)^{x-1}, x = 1, 2, \dots$ and $p(x) = \binom{x-1}{r-1} p^r (1 - p)^{x-r}$ $x = r, r + 1, r + 2, \dots$
	3.2	Mean and variance of a geometric distribution with parameter p .	Use of $\mu = \frac{1}{p}$ and $\sigma^2 = \frac{1-p}{p^2}$
	3.3	Mean and variance of negative binomial distribution with $P(X = x)$ $= \binom{x-1}{r-1} p^r (1 - p)^{(x-r)}$	Use of $\mu = \frac{r}{p}$ and $\sigma^2 = \frac{r(1-p)}{p^2}$